Frequency Distribution

A tabular arrangement of raw data by a certain number of classes and the number of items (called frequency) belonging to each class is termed as a frequency distribution or A grouping of data into categories showing the number of observations in each mutually exclusive category.

Example; Income of person, Numbers in different groups, runs per over, weaves per meter etc.

The frequency distributions are of two types, namely, discrete frequency distribution and continuous frequency distribution.

Discrete Frequency Distribution

Raw data sometimes may contain a limited number of values and each of them appeared many numbers of times. Such data may be organized in a tabular form termed as a simple frequency distribution. Thus the tabular arrangement of the data values along with the frequencies is a simple frequency distribution. A simple frequency distribution is formed using a tool called 'tally chart'. A tally chart is constructed using the following method:

Examine each data value.

Record the occurrence of the value with the slash symbol (/), called tally bar or tally mark.

If the tally marks are more than four, put a crossbar on the four tally bar and make this as block of 5 tally bars (////)

Find the frequency of the data value as the total number of tally bars i.e., tally marks corresponding to that value.

Example (ungrouped data)

The marks obtained by 25 students in a test are given as follows: 10, 20, 20, 30, 40, 25, 25, 30, 40, 20, 25, 25, 50, 15, 25, 30, 40, 50, 40, 50, 30, 25, 25, 15 and 40. The following discrete frequency distribution represents the given data:

Table 3.11

Marks Scored by the Students

Marks	Tally Bars	Frequency
10	/	1
15	//	2
20	///	3
25	~~////~~ //	7
30	////	4
40	~~////~~	5
50	///	3
Total		25

Continuous Frequency Distribution:

It is necessary to summarize and present large masses of data so that important facts from the data could be extracted for effective decisions. A large mass of data that is summarized in such a way that the data values are distributed into groups, or classes, or categories along with the frequencies is known as a continuous or grouped frequency distribution.

Steps of Constructing Frequency Distribution

The following steps can be taken to construct a frequency distribution table:

1. **Finding Total observations :** Firstly, we have to find the Total observations.
2. **Finding Highest and Lowest Values:** Find the highest and the lowest value of the data set.
3. **Finding Range : Range = Highest value - Lowest Value .**
4. **Finding Number of Classes:** Find the number of classes by using

   k = 1 + 3.322 log N.

   Where, N=Total observations
5. **Finding Class Interval:** Find the class interval by using

$$= \frac{Highest\ \text{Value} - \text{Lowest Value}}{1+3.322\log \text{N}} = \frac{Range}{k}$$

6. Write a table with heading **Class interval, Tally marks** and **Frequency.**
7. Start the class with lowest value up to reach highest value and every class has equal interval .
8. Count the value with **Tally Marks** (////) and write respective classes.
9. Find the frequencies for each class by counting the number of tallies of each class.
10. **Find total frequency and write bottom of the third column.**

Example 1: Monthly income (in thousand taka) of 30 families given below (Group data):

20	22	35	42	37	42	48	53	49	65
39	48	67	18	12	23	37	35	49	63
65	55	45	58	57	69	25	29	58	65

Construct a frequency distribution table.

Solution:

1. Finding Total observations: Firstly, Find the Total observations (N=30)

2. Finding Highest and Lowest Values: Find the highest (H) =69 and the lowest value (L) =12 of the data set.

3. Finding Range : Range = Highest value - Lowest Value =69-12 =57 .

4. Finding Number of Classes: Find the number of classes by using

k = 1 + 3.322 log N.

K=1+3-322log30

K=1+3.322x1.477

K=1+4.906=5.906

K=5.906≈ 6

5 .Finding Class Interval: Find the class interval by using

$$= \frac{Highest\ \text{Value} - \text{Lowest Value}}{1+3.322\log \text{N}} = \frac{Range}{k} = \frac{57}{6} = 9.5 \approx 10$$

6. Write a table with heading **Class interval, Tally marks** and **Frequency.**

Frequency distribution table

Class Interval	Tally marks	Frequency
12-22	////	4
22-32	///	3
32-42	~~////~~ //	7
42-52	~~////~~	5
52-62	~~////~~	5
62-72	~~///~~/ /	6
Total		N=30

7. Start the class interval with lowest value up to reach highest value and every class has equal interval and which is 10.

8. Count the value with **Tally Marks** (////) and write respective classes.

9. Find the frequencies for each class by counting the number of tallies of each class.

10. Find total frequency and write bottom of the third column.

Example 2: In a class examination, the marks obtained by 30 students are given below:

~~32~~ ~~33~~ ~~55~~ ~~47~~ ~~21~~ ~~50~~ 27 26 24 33

62 42 38 15 45 32 44 48 68 49

40 52 30 17 44 58 48 39 42 37

Make out a frequency distribution from the above data taking appropriate class interval.

Solution:

1. Finding Total observations: Firstly, Find the Total observations (N=30)

2. Finding Highest and Lowest Values: Find the highest (H) =68 and the lowest value (L) =15 of the data set.

3. Finding Range : Range = Highest value - Lowest Value =68-15 =53 .

4. Finding Number of Classes: Find the number of classes by using

k = 1 + 3.322 log N.

K=1+3-322log30

K=1+3.322x1.477

K=1+4.906=5.906

K=5.906≈6

5 .Finding Class Interval: Find the class interval by using

$$= \frac{Highest\ \text{Value} - \text{Lowest Value}}{1+3.322\log \text{N}} = \frac{Range}{k} = \frac{53}{6} = 8.83 \approx 9$$

6. Write a table with heading **Class interval, Tally marks** and **Frequency.**

Frequency distribution Table

Class Interval	Tally marks	Frequency
15-24	////	4
25-34	~~////~~ //	7
35-44	~~////~~ ///	8
45-54	~~////~~ //	7
55-64	///	3
65-74	/	1
Total		**N=30**

7. Start the class interval with lowest value up to reach highest value and every class has equal interval and which is 10.

8. Count the value with **Tally Marks** (////) and write respective classes.

9. Find the frequencies for each class by counting the number of tallies of each class.

10. Find total frequency and write bottom of the third column.

Problem: The earning of 39 firms for the year 2015--2016 are given below (in lakh taka):

30	42	30	54	40	48	15	17	51	42	35	41	30
27	42	36	28	26	37	54	44	31	36	40	36	22
30	31	19	48	16	42	32	21	22	40	33	41	21

i) Construct a frequency distribution table from above information. or

ii) Construct a frequency distribution table taking a suitable class-interval. or

iii) Construct a frequency distribution table taking continuous class-interval. or

iv) Construct a frequency distribution table taking class interval is 5 or 10 or 14.or

v) Construct a frequency distribution table start with class interval 15-25.or

vi) Construct a frequency distribution with an example (raw data).

Example: In a class examination, the marks obtained by 30 students are given below:

~~32~~ ~~33~~ ~~55~~ ~~47~~ ~~21~~ ~~50~~ 27 26 24 33

62 42 38 15 45 32 44 48 68 49

40   52   30   17   44   58   48   39   42   37

Make out a frequency distribution from the above data taking continuous class intervals (appropriate) class interval.

Frequency distribution table

Class Interval	**Tally marks**	**Frequency**
15-24	**////**	**4**
25-34 35-44	**~~////~~ //**	**7**
45-54	**~~////~~ ///**	**8**
55-64	**~~////~~ //**	**7**
65-74	**///**	**3**
	/	**1**
Total		**N=30**

15-24

Class Interval:

Class Interval	**Class boundary**
15-24	**14.5-24.5**
25-34	**24.5-34.5**
35-44	**34.5-44.5**
45-54	**44.5-54.5**
55-64	**54.5-64.5**
65-74	**64.5-74.5**

Class interval two types i) Discrete class interval: 0-4,5-9,10-14……..

ii) Continuous class interval: 0-5,5-10,10-15……..

Class boundary:

Lower class boundary =lower class limit -1/2(.5)=15-.5=14.5

Upper class boundary =upper class limit+1/2(.5)=24+.5=24.5

Another two types class interval: i) Close class interval: 0-4,5-9,10-14,15-19

ii) Open class interval: 0-4,5-9,10-14,15-19, 20+

Mid value:
Mid value=(upper class limit +lower class limit)/2
For example , 15-24 ,mid value =(15+24)/2=19.5.
Some times mid value is called midpoint or average point.

Class Interval	Mid value (xi)	Frequency
15-24 25-34 35-44 45-54 55-64 65-74	19.5 15+24/2 29.5 39.5 49.5 59.5 69.5	4 7 8 7 3 1
Total		N=30

Frequency:

Class Interval	Frequency(fi)
15-24 25-34 35-44 45-54 55-64 65-74	4(class frequency) 7 8 7 3 1
Total	N=30(Total frequency)

Cumulative frequency (Cf):

Cumulative frequency Two types :i) Less than Cumulative frequency

ii) More than Cumulative frequency

Class Interval	Frequency(fi)	Cumulative Frequency(cf)	
		Less than	More than
15-24	4	4	30(26+4)
25-34	7	11(4+7)	26(19+7)
35-44	8	19(11+8)	19(11+8)
45-54	7	26(19+7)	11(4+7)
55-64	3	29(26+3)	4(1+3)
65-74	1	30(29+1)	1
Total			

Relative frequency (Rf): $\frac{Class\ frequcncy}{Total\ frequency}$**x100**

Class Interval	Frequency(fi)	Relative frequency(Rf)
15-24	4	13.33% ($\frac{4}{30}\times 100$¿
25-34	7	23.33% ($\frac{7}{30}\times 100$¿
35-44	8	

45-54 **55-64** **65-74**	**7** **3** **1**	**26.66%** ($\frac{8}{30}\times 100$ ¿ **23.33%** ($\frac{7}{30}\times 100$ ¿ **10%** ($\frac{3}{30}\times 100$ ¿ **3.33%** ($\frac{4}{30}\times 100$ ¿
Total	**N=30**	

Class Interval	**Mid value (xi)**	**Frequency**	Cumulative frequency(Cf)		Relative frequency(Rf)
			Less than	More than	
15-24	**19.5**	**4**	4	30	13.33%
25-34	**29.5**	**7**	11	26	23.33%
35-44	**39.5**	**8**	19	19	26.66%
45-54	**49.5**	**7**	26	11	23.33%
55-64	**59.5**	**3**	29	4	10.0%
65-74	**69.5**	**1**	30	1	3.33%
Total		**N=30**			

Frequency Density =Frequency of a certain class/class interval
Cumulative relative frequency = class cumulative frequency /total frequency

Questions:

1) What is frequency distribution? Write the different types of frequency distribution.
2) Define the following terms :i) class interval ii) mid value iii) class boundary iv) Frequency v) cumulative frequency and its types vi) relative frequency

Problem:

The earning of 39 firms for the year 2015--2016 are given below (in lakh taka):

30	42	30	54	40	48	15	17	51	42	35	41	30
27	42	36	28	26	37	54	44	31	36	40	36	22
30	31	19	48	16	42	32	21	22	40	33	41	21

i) Construct a frequency distribution table taking continuous class-interval.
ii) **Also find cumulative frequency and relative frequency.**

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